



EAST WEST UNIVERSITY

Department of Mathematical & Physical Sciences

Assignment, Summer 2025 Semester

Course:	PHY 109 : Engineering Physics – I
Instructor:	Md. Mahfuzul Haque, Assistant Professor, Department of MPS

1. Imagine a Carnot engine that operates between the temperatures $T_H = 850$ K and $T_L = 300$ K. The engine performs 1200 J of work each cycle, which takes 0.25 s.
(a) What is the efficiency of this engine? (b) How much energy $|Q_H|$ is extracted as heat from the high-temperature reservoir every cycle? (c) How much energy $|Q_L|$ is delivered as heat to the low temperature reservoir every cycle?
2. In Figure 2, the charges are $q_1 = 1.60 \times 10^{-19}$ C and $q_2 = 3.20 \times 10^{-19}$ C, and the separation between them is $R = 0.0200$ m. Charge $q_3 = -3.20 \times 10^{-19}$ C is at a distance $3/4R$ from particle 1. What is the net electrostatic force on particle 1 due to particles 2 and 3?

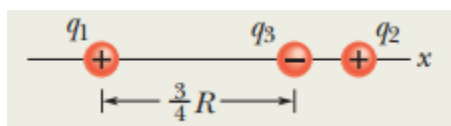


Figure 2

3. In Figure 3, two point charges, $q_1 = +Q$ and $q_2 = +4Q$, are placed along a straight line and separated by a distance $L = 10$ cm. Determine the point along the line, where the net electric field becomes zero.

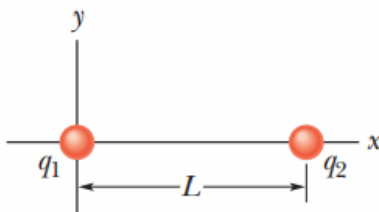


Figure 3

4. An infinitely long straight wire carries a uniform linear charge density $\lambda = 2.0 \times 10^{-8} \text{ C/m}$. Find the magnitude of the electric field at a point located $r = 0.05 \text{ m}$ from the wire. (Take $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$)
5. Figure 5 shows a Gaussian surface in the form of a closed cylinder (a Gaussian cylinder or G-cylinder) of radius R . It lies in a uniform electric field $\vec{E}$ with the cylinder's central axis (along the length of the cylinder) parallel to the field. What is the net flux Φ of the electric field through the cylinder?

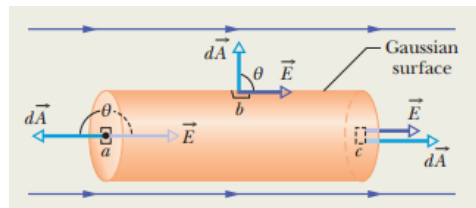


Figure 5

6. Figure 6 shows three situations in which a charged particle with velocity $\vec{v}$ travels through a uniform magnetic field $\vec{B}$. In each situation, what is the direction of the magnetic force $\vec{F}_B$ on the particle?

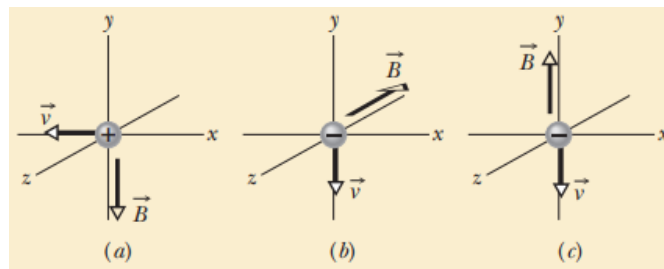


Figure 6

7. A uniform magnetic field with magnitude 1.2 mT , is directed vertically upward throughout the volume of a laboratory chamber. A proton with velocity $3.2 \times 10^7 \text{ m/s}$ enters the chamber, moving horizontally from south to north. What magnetic deflecting force acts on the proton as it enters the chamber? (The proton mass is $1.67 \times 10^{-27} \text{ kg}$.)
8. Figure 8 shows a current i through a wire in a uniform magnetic field $\vec{B}$, as well as the magnetic force $\vec{F}_B$ acting on the wire. In what direction is the field?

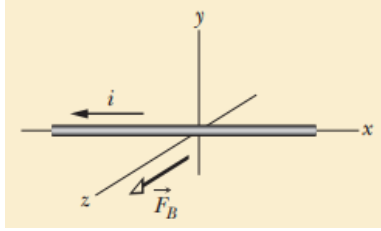


Figure 8

9. Consider two long, parallel, current-carrying wires placed close to each other. Using the right-hand rule and the concept of magnetic fields around a current-carrying conductor, determine the direction of the magnetic force acting on one wire due to the other for the following cases: a. Both wires carry current in the same direction. b. The two wires carry current in opposite directions.
10. In Figure 10, two point charges, $q_1 = +10\text{C}$ and $q_2 = +40\text{C}$, are placed along a straight line and separated by a distance $L = 10\text{ cm}$. Determine the electric field strength at the center point of the line.

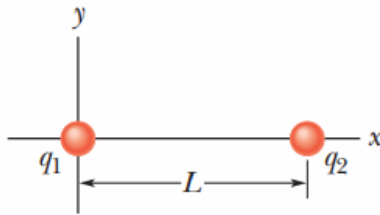


Figure 10